

Priyanka D

Software Engineer

Traverse City, MI | +1 (231) 486 5398 | priyachowdary8.se@gmail.com | [LinkedIn](#)

SUMMARY

Software Engineer with nearly **3+ years** of hands-on experience building scalable APIs and distributed systems using Python (FastAPI) and Java (Spring Boot). Worked on high-volume payment systems at PhonePe, designing resilient backend services with idempotent APIs, Kafka-based asynchronous workflows, and robust failure handling. Also experienced in full-stack development (React, Node.js) and AI-integrated product platforms with a strong focus on performance, reliability and real-world business workflows.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript (ES6+), SQL

Backend Development: FastAPI, Spring Boot, Node.js, Express.js, RESTful API Design, Service-Oriented Architecture

Frontend Development: React.js, HTML5, CSS3

Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis

Distributed Systems & Messaging: Apache Kafka, Event-Driven Architecture, Asynchronous Messaging, Idempotent API Design, Transaction State Management

Authentication & Security: JWT (JSON Web Tokens), Role-Based Access Control (RBAC), Service-to-Service Authentication

Cloud & DevOps: AWS (RDS, S3), Azure (Azure DevOps, Managed Databases), Docker, CI/CD Pipelines, Git, Linux

Testing & Quality: PyTest, Unit Testing, API Testing, Code Reviews, Debugging, Production Incident Support

Core Engineering Concepts: Scalable Backend Design, API Performance Optimization, Concurrency Handling, Failure Recovery, Data Modeling, Agile/Scrum

PROFESSIONAL EXPERIENCE

Starteryou | Software Engineer Intern (Full Stack)

Jan 2025 - May 2025 | USA

- Developed full-stack features using React.js and Node.js, supporting user onboarding, profile creation, and job-application workflows for a career discovery platform.
- Built and maintained RESTful APIs powering core product flows and integrating with AI-driven job-matching services, consuming recommendation outputs to enable personalized job suggestions and one-click applications.
- Worked closely with Python (Django)-based AI services, supporting data pipelines that transformed student and employer information into recommendation results used in live user experiences.
- Designed and optimized MongoDB and PostgreSQL data models, working with AWS RDS (PostgreSQL) for structured job application tracking alongside flexible user profile data.
- Implemented JWT-based authentication and role-based access control (RBAC), securing APIs across student, employer, and admin user roles.
- Improved API performance and reliability by refining database queries and response payloads, ensuring smooth handling of data-heavy workflows under concurrent usage.
- Supported Docker-based deployments to AWS-hosted environments and CI/CD pipelines, contributing to structured release and deployment processes.
- Integrated backend services with Amazon S3 for file and document storage used in profile and job application workflows.
- Collaborated with senior engineers through code reviews and debugging sessions, delivering production-ready fixes and improving system stability during feature rollouts.

PhonePe | Software Engineer II

Jun 2022 - Dec 2023 | India

- Developed and enhanced backend services using Java (Spring Boot) and Python (FastAPI) for UPI payment workflows, handling transaction initiation, retries, failure scenarios, and final state updates in high-volume production systems.
- Built idempotent REST APIs to safely process duplicate client requests and bank callbacks, preventing double debits and inconsistent transaction states.
- Contributed to implementing transaction state machines (INITIATED, PROCESSING, SUCCESS, FAILED, REVERSED) and supported asynchronous processing with Kafka for retries and delayed bank responses.
- Designed and optimized MySQL queries for transaction, audit, and settlement tables, improving reconciliation performance and reducing settlement delays during peak load periods.
- Used Redis for idempotency key storage and short-lived caching, ensuring consistent behavior during retries and transient network failures.
- Implemented service-to-service authentication (JWT) and RBAC, securing internal payment and audit APIs.
- Investigated and resolved P0/P1 production incidents using structured logging, metrics, and distributed tracing, performing root-cause analysis and contributing fixes to improve system reliability.
- Supported Docker-based deployments to Azure-hosted environments, using Azure DevOps CI/CD pipelines to promote services across dev, staging, and production, and assisting with monitored rollouts and rollback procedures.
- Collaborated with product, risk, and operations teams to address edge cases, regulatory requirements, and real-world payment failure patterns.
- Worked with Azure-managed MySQL databases for transactional services, collaborating with platform teams on schema changes, performance tuning, and production troubleshooting.

LTIMindtree | Full-Stack Developer

Mar 2021 - May 2022 | India

- Developed client-facing web applications using React.js and Python (FastAPI), building REST APIs for data retrieval and business workflows, improving application performance by 18% through optimized API responses and reusable UI components.
- Contributed to FastAPI-based backend services, implementing request validation (Pydantic), routing, service logic, and database integrations under senior guidance, reducing API-related defects by 22% across releases.
- Assisted in maintaining a legacy Django-based internal module, fixing bugs and updating serializers and models, gaining exposure to Django ORM and admin workflows.
- Wrote and optimized SQL queries in PostgreSQL for reporting and operational endpoints, working with AWS RDS (PostgreSQL) to improve query execution time by 25% for frequently accessed workflows.
- Implemented JWT-based authentication and supported role-based access logic for secure API consumption across user types.
- Supported Docker-based deployments to AWS-hosted environments and assisted with CI/CD pipelines, reducing environment-related issues and improving build consistency.
- Delivered 30+ Jira stories in an Agile Scrum environment, covering UI enhancements, backend API fixes, and minor feature implementations.
- Collaborated with QA and senior engineers on unit testing (PyTest) and code reviews, contributing to a 20% reduction in post-release defects through early issue detection.
- Supported MongoDB schema updates alongside relational PostgreSQL models to accommodate evolving reporting and application requirements.

EDUCATION

Master of Science - Information Technology

Clark University, Worcester, MA

Jan 2024 - Dec-2025

Bachelor of Science - Computer Science

Ballari Institute of Technology and Management, India

Aug 2019 - Aug 2023

CERTIFICATIONS

Java J2EE - Wipro Talent Next Digital (2022)

Web Technologies - QSpiders (2023)

Microsoft Azure Fundamentals (AZ-900) -Microsoft (2025)

PROJECTS

Decentralized File Storage System

- Built a decentralized file storage platform where files are uploaded to IPFS and only their CIDs and metadata are stored on chain, avoiding blockchain bloat.
- Wrote Solidity smart contracts to persist file hashes, uploader wallet address, timestamps, and permission mappings for read access.
- Implemented a TypeScript frontend that connects to MetaMask, handles wallet state, and signs transactions for file registration.
- Integrated Web3.js to bind frontend state to smart contract calls and emitted events, ensuring file ownership and permissions were always derived from the blockchain.
- Built the storage flow around Pinata IPFS, generating and persisting CIDs per upload and resolving files from public gateways without routing content through the backend.
- Used Hardhat to spin up a local blockchain, write repeatable contract tests, and deploy versioned contracts to Ethereum test nets during development.
- Enforced file authenticity by comparing on-chain CID references with retrieved IPFS hashes, eliminating trust in server-side file metadata.

Artist Portfolio and E-commerce Platform

- Built a MERN-based platform where artists upload, manage, and sell artwork, while buyers browse, choose their favorite, and purchase items.
- Implemented JWT-based session handling and OAuth authentication, enforcing strict role separation that controlled artist-only actions such as uploads and pricing, and buyer-only flows such as purchases and order history.
- Designed REST APIs for artwork CRUD, cart management, order lifecycle, and payment processing, with MongoDB schema optimization.
- Integrated an embedding-based recommendation engine to surface visually and contextually similar artworks based on user interaction history.
- Built artwork discovery features including category-based search, price filtering, popularity sorting, and pagination to support fast browsing at scale.
- Implemented a secure checkout and payment workflow with server-side validation, order creation, transaction state management, and post-payment status updates.
- Added favourites and user interaction tracking, persisting behavioural signals that power engagement metrics and drive the artwork recommendation pipeline.